## 4.8 Other Discrete Probability Distributions

### 4.8.1 The Geometric Random Variable

Suppose that independent trials, each having a probability p,0<p<1, of being a success, are performed until a success occurs. If we let X equal the number of trials required, then

$$P\{X=n\}=(1-p)^{n-1}p \quad n=1,2,\ldots \tag{8.1}$$

**Equation (8.1)** follows because, in order for X to equal n, it is necessary and sufficient that the first n−1 trials are failures and the nth trial is a success. **Equation (8.1)** then follows, since the outcomes of the successive trials are assumed to be independent.

Since

$$\sum_{n=1}^{\infty}P\{X=n\}=p\sum_{n=1}^{\infty}(1-p)^{n-1}=p\frac{1}{1-(1-p)}=1$$

it follows that with probability 1, a success will eventually occur. Any random variable X whose probability mass function is given by **Equation (8.1)** is said to be a *geometric* random variable with parameter p.

***Example 8a***

An urn contains N white and M black balls. Balls are randomly selected, one at a time, until a black one is obtained. If we assume that each ball selected is replaced before the next one is drawn, what is the probability that

a. exactly n draws are needed?
b. at least k draws are needed?

Solution

If we let X denote the number of draws needed to select a black ball, then X satisfies **Equation (8.1)** with p=M/(M+N). Hence,

a. $P\{X=n\}=(\frac{N}{M+N})^{n-1}\frac{M}{M+N}=\frac{MN^{n-1}}{(M+N)^n}$

b. $P\{X\geq k\}=\frac{M}{M+N}\sum_{n=k}^{\infty}(\frac{N}{M+N})^{n-1}=(\frac{M}{M+N})(\frac{N}{M+N})^{k-1}/[1-\frac{N}{M+N}]=(\frac{N}{M+N})^{k-1}$

Of course, part (b) could have been obtained directly, since the probability that at least k trials are necessary to obtain a success is equal to the probability that the first k−1 trials are all failures. That is, for a geometric random variable,

$$P\{X\geq k\}=(1-p)^{k-1}$$

***Example 8b***

Find the expected value of a geometric random variable.

Solution

With q=1−p, we have

$$E[X]=\sum_{i=1}^{\infty}iq^{i-1}p=\sum_{i=1}^{\infty}(i-1+1)q^{i-1}p=\sum_{i=1}^{\infty}(i-1)q^{i-1}p+\sum_{i=1}^{\infty}q^{i-1}p=\sum_{i=1}^{\infty}jq^{j}p+1=q\sum_{j=1}^{\infty}jq^{j-1}p+1=qE[X]+1$$

Hence,

$$pE[X]=1$$

yielding the result

$$E[X]=\frac{1}{p}$$

In other words, if independent trials having a common probability p of being successful are performed until the first success occurs, then the expected number of required trials equals 1/p. For instance, the expected number of rolls of a fair die that it takes to obtain the value 1 is 6.

***Example 8c***

Find the variance of a geometric random variable.

Solution

To determine Var(X), let us first compute $E[X^2]$. With q=1−p, we have

$$E[X^2]=\sum_{i=1}^{\infty}i^2q^{i-1}p=\sum_{i=1}^{\infty}(i-1+1)^2q^{i-1}p=\sum_{i=1}^{\infty}(i-1)^2q^{i-1}p+\sum_{i=1}^{\infty}2(i-1)q^{i-1}p+\sum_{i=1}^{\infty}q^{i-1}p=\sum_{i=1}^{\infty}j^2q^{j}p+2\sum_{j=1}^{\infty}jq^{j}p+1=qE[X^2]+2qE[X]+1$$

Using E[X]=1/p, the equation for $E[X^2]$ yields

$$pE[X^2]=2qp+1$$

Hence,

$$E[X^2]=\frac{2q}{p}+p\frac{p^2}{}=\frac{q+1}{p^2}$$

giving the result

$$Var(X)=\frac{q+1}{p^2}-\frac{1}{p^2}=\frac{q}{p^2}=\frac{1-p}{p^2}$$

### 4.8.2 The Negative Binomial Random Variable

Suppose that independent trials, each having probability p,0<p<1, of being a success are performed until a total of r successes is accumulated. If we let X equal the number of trials required, then

$$P\{X=n\}=\binom{n-1}{r-1}p^r(1-p)^{n-r} \quad n=r,r+1,\ldots \tag{8.2}$$

**Equation (8.2)** follows because, in order for the rth success to occur at the nth trial, there must be r−1 successes in the first n−1 trials and the nth trial must be a success. The probability of the first event is

$$\binom{n-1}{r-1}p^{r-1}(1-p)^{n-r}$$

and the probability of the second is p; thus, by independence, **Equation (8.2)** is established. To verify that a total of r successes must eventually be accumulated, either we can prove analytically that

$$\sum_{n=r}^{\infty}P\{X=n\}=\sum_{n=r}^{\infty}\binom{n-1}{r-1}p^r(1-p)^{n-r}=1 \tag{8.3}$$

or we can give a probabilistic argument as follows: The number of trials required to obtain r successes can be expressed as $Y_1+Y_2+\cdots+Y_r$, where $Y_1$ equals the number of trials required for the first success, $Y_2$ the number of additional trials after the first success until the second success

occurs, Y3 the number of additional trials until the third success, and so on. Because the trials are independent and all have the same probability of success, it follows that Y1,Y2,…,Yr are all geometric random variables. Hence, each is finite with probability 1, so ∑i=1rYi must also be finite, establishing **Equation (8.3)**.

Any random variable X whose probability mass function is given by **Equation (8.2)** is said to be a *negative binomial* random variable with parameters (r, p). Note that a geometric random variable is just a negative binomial with parameter (1, p).

In the next example, we use the negative binomial to obtain another solution of the problem of the points.

***Example 8d***

If independent trials, each resulting in a success with probability p, are performed, what is the probability of r successes occurring before s failures?

Solution

The solution will be arrived at by noting that r successes will occur before s failures if and only if the rth success occurs no later than the (r+s−1) trial. This follows because if the rth success occurs before or at the (r+s−1) trial, then it must have occurred before the sth failure, and conversely. Hence, from **Equation (8.2)**, the desired probability is

$$\sum_{n=r}^{r+s-1}\binom{n-1}{r-1}p^r(1-p)^{n-r}$$

***Example 8e The Banach match problem***

At all times, a pipe-smoking mathematician carries 2 matchboxes—1 in his left-hand pocket and 1 in his right-hand pocket. Each time he needs a match, he is equally likely to take it from either pocket. Consider the moment when the mathematician first discovers that one of his matchboxes is empty. If it is assumed that both matchboxes initially contained N matches, what is the probability that there are exactly k matches, k=0,1,…,N, in the other box?

Solution

Let E denote the event that the mathematician first discovers that the right-hand matchbox is empty and that there are k matches in the left-hand box at the time. Now, this event will occur if and only if the (N+1) choice of the right-hand matchbox is made at the (N+1+N−k) trial. Hence, from **Equation (8.2)** (with p=12,r=N+1,and n=2N−k+1), we see that

$$P(E)=\binom{2N-k}{N}\left(\frac{1}{2}\right)^{2N-k+1}$$

Since there is an equal probability that it is the left-hand box that is first discovered to be empty and there are k matches in the right-hand box at that time, the desired result is

$$2P(E)=\binom{2N-k}{N}\left(\frac{1}{2}\right)^{2N-k}$$

***Example 8f***

Compute the expected value and the variance of a negative binomial random variable with parameters r and p.

Solution

We have

$$E[X^k]=\sum_{n=r}^{\infty}n^k\binom{n-1}{r-1}p^r(1-p)^{n-r}=\frac{r}{p}\sum_{n=r}^{\infty}n^k\binom{n-1}{r}p^{r+1}(1-p)^{n-r} \text{ since } n\binom{n-1}{r-1}=r\binom{n}{r}=\frac{r}{p}\sum_{m=r+1}^{\infty}(m-1)^k\binom{m-1}{r}p^{r+1}(1-p)^{m-(r+1)} \text{ by setting } m=n+1$$
$$=\frac{r}{p}E[(Y-1)^k]$$

where Y is a negative binomial random variable with parameters r+1,p. Setting k=1 in the preceding equation yields

$$E[X]=\frac{r}{p}$$

Setting k=2 in the equation for E[Xk] and using the formula for the expected value of a negative binomial random variable gives

$$E[X^2]=\frac{r}{p}E[Y-1]=\frac{r}{p}\left(\frac{r+1}{p}-1\right)$$

Therefore,

$$Var(X)=\frac{r}{p}\left(\frac{r+1}{p}-1\right)-\left(\frac{r}{p}\right)^2=\frac{r(1-p)}{p^2}$$

Thus, from **Example 8f**, if independent trials, each of which is a success with probability p, are performed, then the expected value and variance of the number of trials that it takes to amass r successes is *r/p* and r(1−p)/p2, respectively.

Since a geometric random variable is just a negative binomial with parameter r=1, it follows from the preceding example that the variance of a geometric random variable with parameter p is equal to (1−p)/p2, which checks with the result of **Example 8c**.

***Example 8g***

Find the expected value and the variance of the number of times one must throw a die until the outcome 1 has occurred 4 times.

Solution

Since the random variable of interest is a negative binomial with parameters r=4 and p=16, it follows that

$$E[X]=24 \quad Var(X)=4(56)(16)^2=120$$

Now, let us suppose that the independent trials are not ended when there have been a total of r successes, but that they continue on. Aside from X, the number of trials until there have been r successes, some other random variables of interest are, for s>0,

*Y* : the number of trials until there have been s failures;

*V* : the number of trials until there have been either r successes or s failures;

*Z* : the number of trials until there have been both at least r successes and at least s failures.

Because each trial is independently a failure with probability 1−p, it follows that Y is a negative binomial random variable with probability mass function

$$P(Y=n)=\binom{n-1}{s-1}(1-p)^s p^{n-s}, \quad n\geq s$$

To determine the probability mass function of V=min(X,Y), note that the possible values of V are all less than r+s. Suppose n<r+s. If either the rth success or the sth failure occurs at time n then, because n<r+s, the other event would not yet have occurred. Consequently, V will equal n if either X or Y is equal to n. Because we cannot have both that X=n and that Y=n, this yields

$$P(V=n)=P(X=n)+P(Y=n)=\binom{n-1}{r-1}p^r(1-p)^{n-r}+\binom{n-1}{s-1}(1-p)^s p^{n-s}, \quad n<r+s$$

To determine the probability mass function of Z=max(X,Y), note that Z≥r+s. For n≥r+s, if either the rth success or the sth failure occurs at time n then the other event must have already occurred by time n. Consequently, for n≥r+s, Z will equal n if either X or Y is equal to n. This gives

$$P(Z=n)=P(X=n)+P(Y=n)=\binom{n-1}{r-1}p^r(1-p)^{n-r}+\binom{n-1}{s-1}(1-p)^s p^{n-s}, \quad n<r+s$$

### 4.8.3 The Hypergeometric Random Variable

Suppose that a sample of size n is to be chosen randomly (without replacement) from an urn containing N balls, of which m are white and N−m are black. If we let X denote the number of white balls selected, then

$$P\{ X=i \}=(mi)(N−mn−i)(Nn)i=0,1,…,n \tag{8.4}$$

A random variable X whose probability mass function is given by **Equation (8.4)** for some values of n, N, m is said to be a *hypergeometric* random variable.

**Remark** Although we have written the hypergeometric probability mass function with i going from 0 to n,P{X=i} will actually be 0, unless i satisfies the inequalities n−(N−m)≤i≤min(n,m). However, **Equation (8.4)** is always valid because of our convention that (rk) is equal to 0 when either k<0 or r<k.

***Example 8h***

An unknown number, say, N, of animals inhabit a certain region. To obtain some information about the size of the population, ecologists often perform the following experiment: They first catch a number, say, m, of these animals, mark them in some manner, and release them. After allowing the marked animals time to disperse throughout the region, a new catch of size, say, n, is made. Let X denote the number of marked animals in this second capture. If we assume that the population of animals in the region remained fixed between the time of the two catches and that each time an animal was caught it was equally likely to be any of the remaining uncaught animals, it follows that X is a hypergeometric random variable such that

$$P\{ X=i \}=(mi)(N−mn−i)(Nn)≡Pi(N)$$

Suppose now that X is observed to equal i. Then, since Pi(N) represents the probability of the observed event when there are actually N animals present in the region, it would appear that a reasonable estimate of N would be the value of N that maximizes Pi(N). Such an estimate is called a *maximum likelihood* estimate. (See **Theoretical Exercises 13** and **18** for other examples of this type of estimation procedure.)

The maximization of Pi(N) can be done most simply by first noting that

$$Pi(N)Pi(N−1)=(N−m)(N−n)N(N−m−n+i)$$

Now, the preceding ratio is greater than 1 if and only if

$$(N−m)(N−n)≥N(N−m−n+i)$$

or, equivalently, if and only if

$$N≤mni$$

Thus, Pi(N) is first increasing and then decreasing and reaches its maximum value at the largest integral value not exceeding *mn/i*. This value is the maximum likelihood estimate of N. For example, suppose that the initial catch consisted of m=50 animals, which are marked and then released. If a subsequent catch consists of n=40 animals of which i=4 are marked, then we would estimate that there are some 500 animals in the region. (Note that the preceding estimate could also have been obtained by assuming that the proportion of marked animals in the region, *m/N*, is approximately equal to the proportion of marked animals in our second catch, *i/n*.)

***Example 8i***

A purchaser of electrical components buys them in lots of size 10. It is his policy to inspect 3 components randomly from a lot and to accept the lot only if all 3 are nondefective. If 30 percent of the lots have 4 defective components and 70 percent have only 1, what proportion of lots does the purchaser reject?

Solution

Let A denote the event that the purchaser accepts a lot. Now,

$$P(A)=P(A|\ \text{lot has 4 defectives})310+P(A\mid \text{lot has 1 defective})710=(40)(63)(103)(310)+(10)(93)(103)(710)=54100$$

Hence, 46 percent of the lots are rejected.

If n balls are randomly chosen without replacement from a set of N balls of which the fraction p=m/N is white, then the number of white balls selected is hypergeometric. Now, it would seem that when m and N are large in relation to n, it shouldn't make much difference whether the selection is being done with or without replacement, because, no matter which balls have previously been selected, when m and N are large, each additional selection will be white with a probability approximately equal to p. In other words, it seems intuitive that when m and N are large in relation to n, the probability mass function of X should approximately be that of a binomial random variable with parameters n and p. To verify this intuition, note that if X is hypergeometric, then, for i≤n,

$$P\{ X=i \}=(mi)(N−mn−i)(Nn)=m!(m−i)!\ i!(N−m)!(N−m−n+i)!\ (n−i)!(N−n)!\ n!N!=(ni)mNm−1N−1⋯m−i+1N−i+1N−mN−iN−m−1N−1−i−1⋯N−m−(n−i−1)N−i−(n−i−1)≈(ni)pi(1−p)n−i$$

when p=m/N and m and N arelarge in relation to n and i

***Example 8j***

Determine the expected value and the variance of X, a hypergeometric random variable with parameters n, N, and m.

Solution

$$E[ Xk ]=∑i=0nikP\{ X=i \}=∑i=1nik(mi)(N−mn−i)/(Nn)$$

Using the identities

$$i(mi)=m(m−1i−1) \text{ and } n(Nn)=N(N−1n−1)$$

we obtain

$$E[ Xk ]=nmN∑i=1nik−1(m−1i−1)(N−mn−i)/(N−1n−1)=nmN∑j=0n−1(j+1)k−1(m−1j)(N−mn−1−j)/(N−1n−1)=nmNE[ (Y+1)k−1 ]$$

where Y is a hypergeometric random variable with parameters n−1, N−1, and m−1. Hence, upon setting k=1, we have

$$E[X]=nmN$$

In words, if n balls are randomly selected from a set of N balls, of which m are white, then the expected number of white balls selected is *nm/N*.

Upon setting k=2 in the equation for E[Xk], we obtain

$$E[ X2 ]=nmNE[ Y+1 ]=nmN[ (n−1)(m−1)N−1+1 ]$$

where the final equality uses our preceding result to compute the expected value of the hypergeometric random variable Y.

Because E[X]=nm/N, we can conclude that

$$Var(X)=nmN[(n−1)(m−1)N−1+1−nmN]$$

Letting p=m/N and using the identity

$$\frac{m-1}{N-1}=\frac{Np-1}{N-1}=\frac{p-\frac{1}{N}}{1-\frac{1}{N}}$$

shows that

$$\mathrm{Var}(X)=np\left[\frac{(n-1)p-(n-1)\frac{1}{N}}{1-\frac{1}{N}}+1-np\right]=np(1-p)\left(\frac{1-\frac{n-1}{N-1}}{1-\frac{1}{N}}\right)$$

**Remark** We have shown in **Example 8j** that if n balls are randomly selected without replacement from a set of N balls, of which the fraction p are white, then the expected number of white balls chosen is *np*. In addition, if N is large in relation to n [so that (N−n)/(N−1) is approximately equal to 1], then

$$\mathrm{Var}(X)\approx np(1-p)$$

In other words, E[ X ] is the same as when the selection of the balls is done with replacement (so that the number of white balls is binomial with parameters n and p), and if the total collection of balls is large, then Var(X) is approximately equal to what it would be if the selection were done with replacement. This is, of course, exactly what we would have guessed, given our earlier result that when the number of balls in the urn is large, the number of white balls chosen approximately has the mass function of a binomial random variable.

### 4.8.4 The Zeta (or Zipf) Distribution

A random variable is said to have a zeta (sometimes called the Zipf) distribution if its probability mass function is given by

$$P\{X=k\}=\frac{C}{k^{\alpha+1}} \quad k=1,2,\ldots$$

for some value of $\alpha>0$. Since the sum of the foregoing probabilities must equal 1, it follows that

$$C=\left[\sum_{k=1}^{\infty}\frac{1}{k^{\alpha+1}}\right]^{-1}$$

The zeta distribution owes its name to the fact that the function

$$\zeta(s)=1+\frac{1}{2^s}+\frac{1}{3^s}+\cdots+\frac{1}{k^s}+\cdots$$

is known in mathematical disciplines as the Riemann zeta function (after the German mathematician G. F. B. Riemann).

The zeta distribution was used by the Italian economist V. Pareto to describe the distribution of family incomes in a given country. However, it was G. K. Zipf who applied zeta distribution to a wide variety of problems in different areas and, in doing so, popularized its use.